Figure 1.

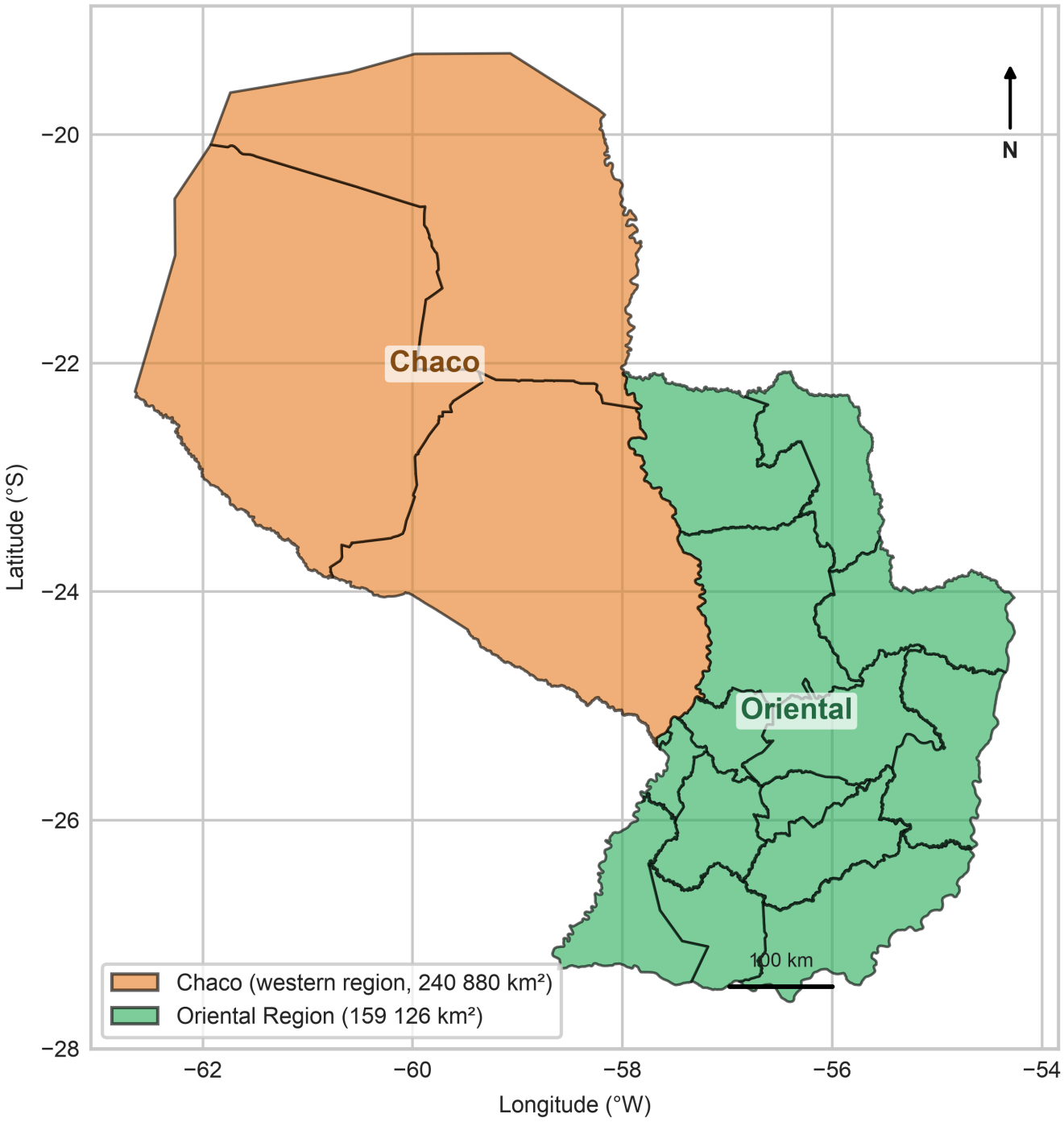


Figure 1. Study domain and regional split: Dry Chaco (west) and Oriental Region (east) over the national 1-km² grid (EPSG:32721).

Figure 2.

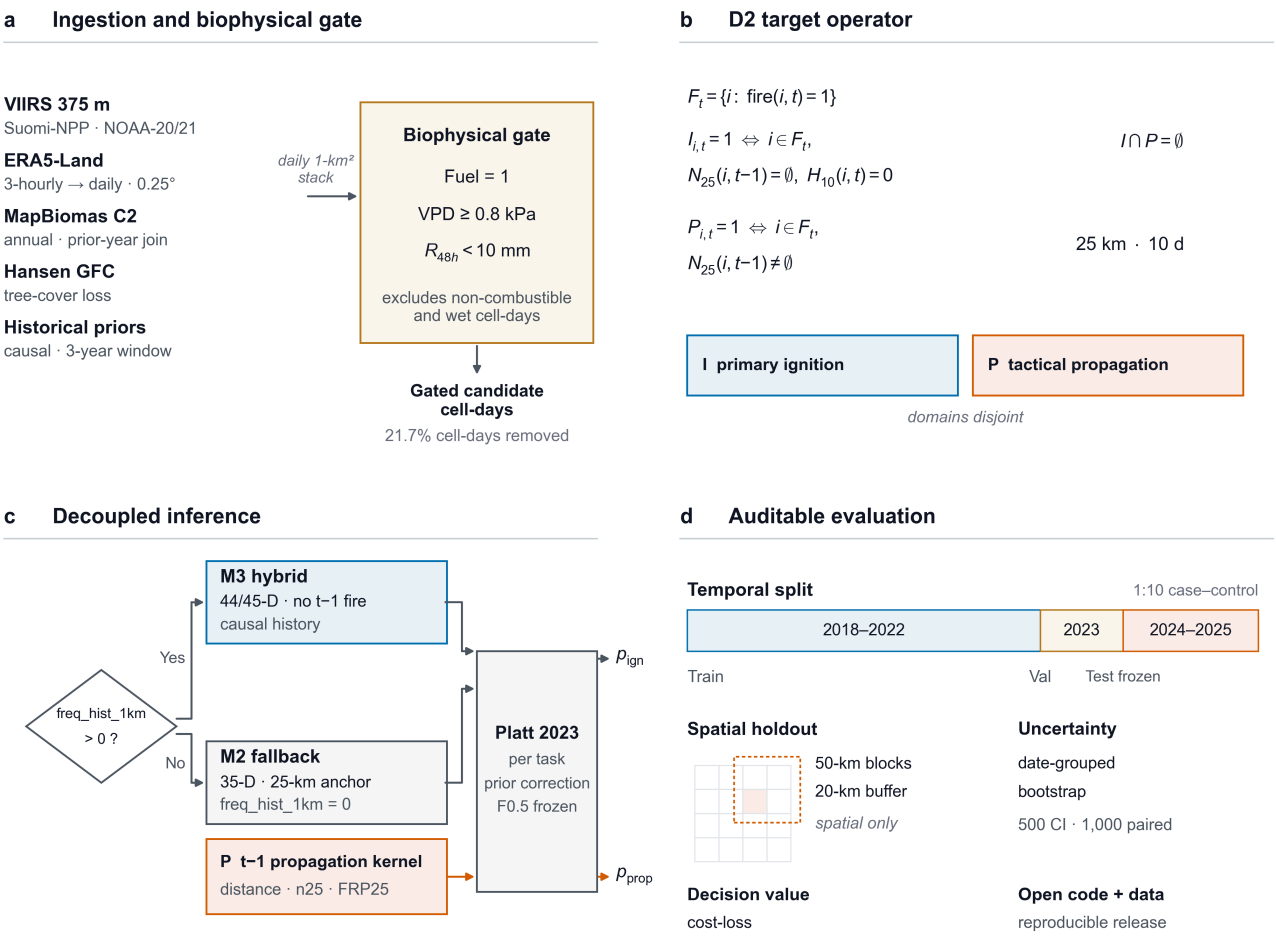


Figure 2. Computational framework. (a) Daily 1-km² inputs (VIIRS 375 m, ERA5-Land 3-hourly, MapBiomas C2, Hansen GFC and causal historical priors) pass through a biophysical gate (R48h < 10 mm is 48-h rainfall; fuel = 1, VPD ≥ 0.8 kPa); the gate removes 21.7 % of cell-days but retains 97.17 % of thermal detections and is not a label filter. (b) D2 uses N25(i,t−1) for the set of cells within 25 km with fire at t−1 and H10(i,t) for any same-cell detection in the previous 10 d; the domains are disjoint. Same-cell 2–10 d recurrences without a t−1 neighbour remain candidate negatives. (c) The 1:10 case-control sample is used for ignition and population denominators are retained. M3 is used where freq_hist_1km > 0 and M2 where it is 0; propagation uses a causal t−1 kernel. Probabilities are calibrated with a per-task monotonic Platt transform fitted on 2023 and frozen; the F0.5 cutoff is applied after prior correction. M1 (24 variables) is a diagnostic baseline and is not shown. (d) Evaluation uses a frozen temporal split (2018–2022/2023/2024–2025), 50-km blocks with a 20-km buffer, date-grouped bootstrap (500 AUC CI; 1,000 paired), cost-loss and open code/data. The 25-km radius is the D2 target; the 20-km buffer applies only to the spatial holdout. Endpoints are ROC-AUC, PR-AUC, Brier, ECE, conformal coverage, population precision/recall and lift; code/data, DOI and hashes are archived.

Figure 3.

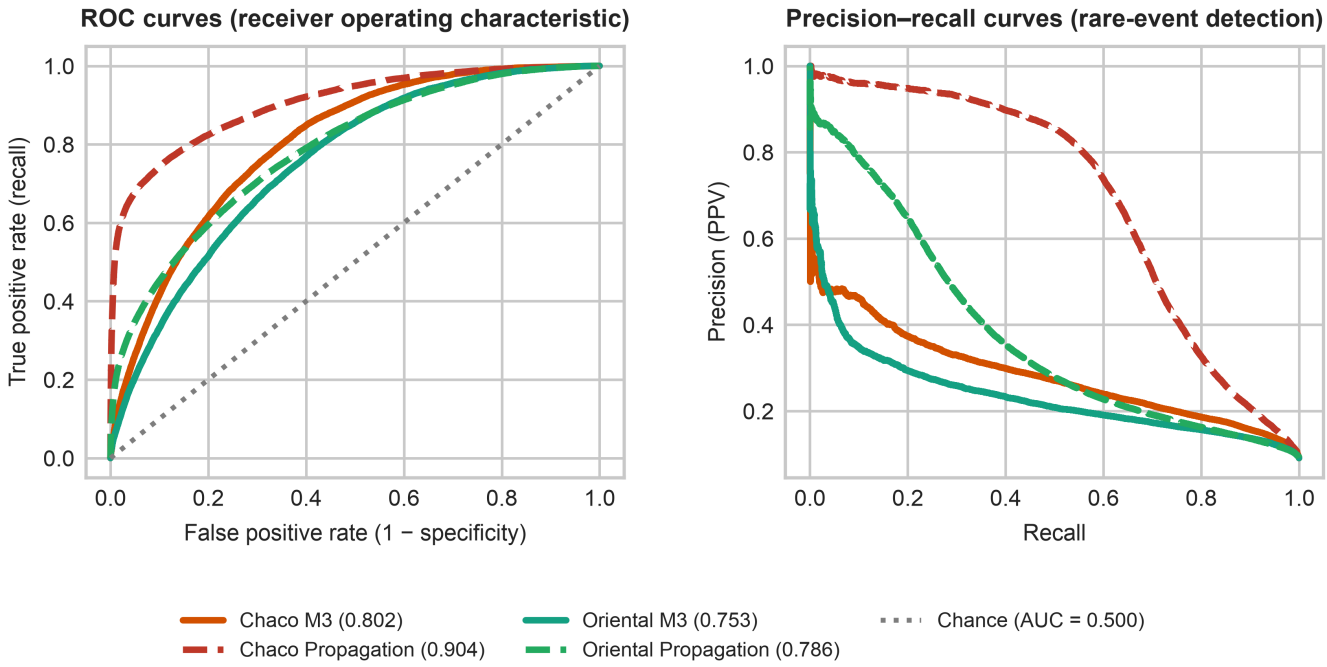


Figure 3. ROC and precision–recall curves for the retrospective D2 test (2024–2025), by region and task. The ignition curves correspond to M3; propagation uses the causal t-1 layer.

Figure 4.

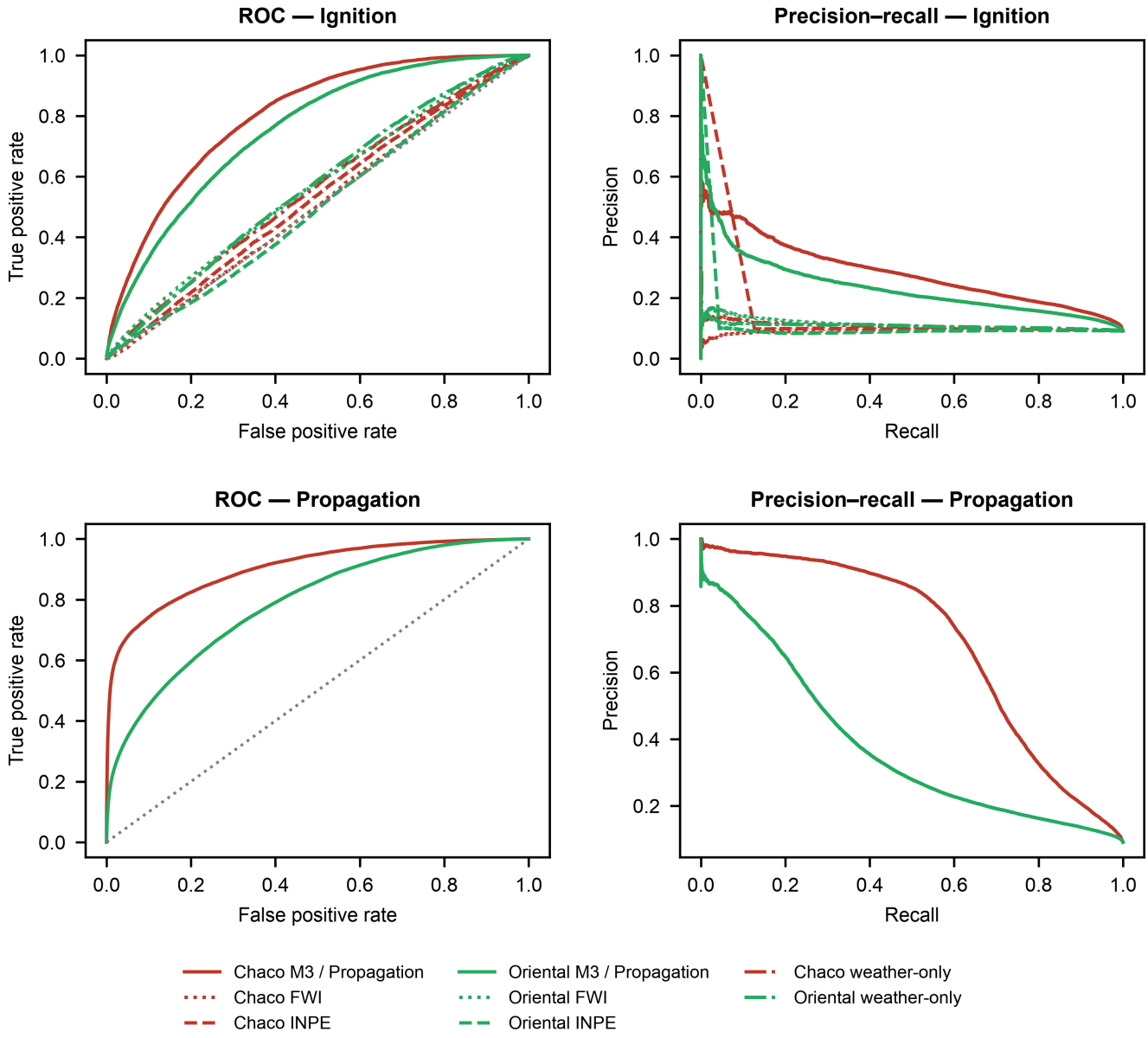


Figure 4. ROC and precision–recall curves for M3 against full Canadian FWI, INPE and the weather-only GBT under D2.

Figure 5.

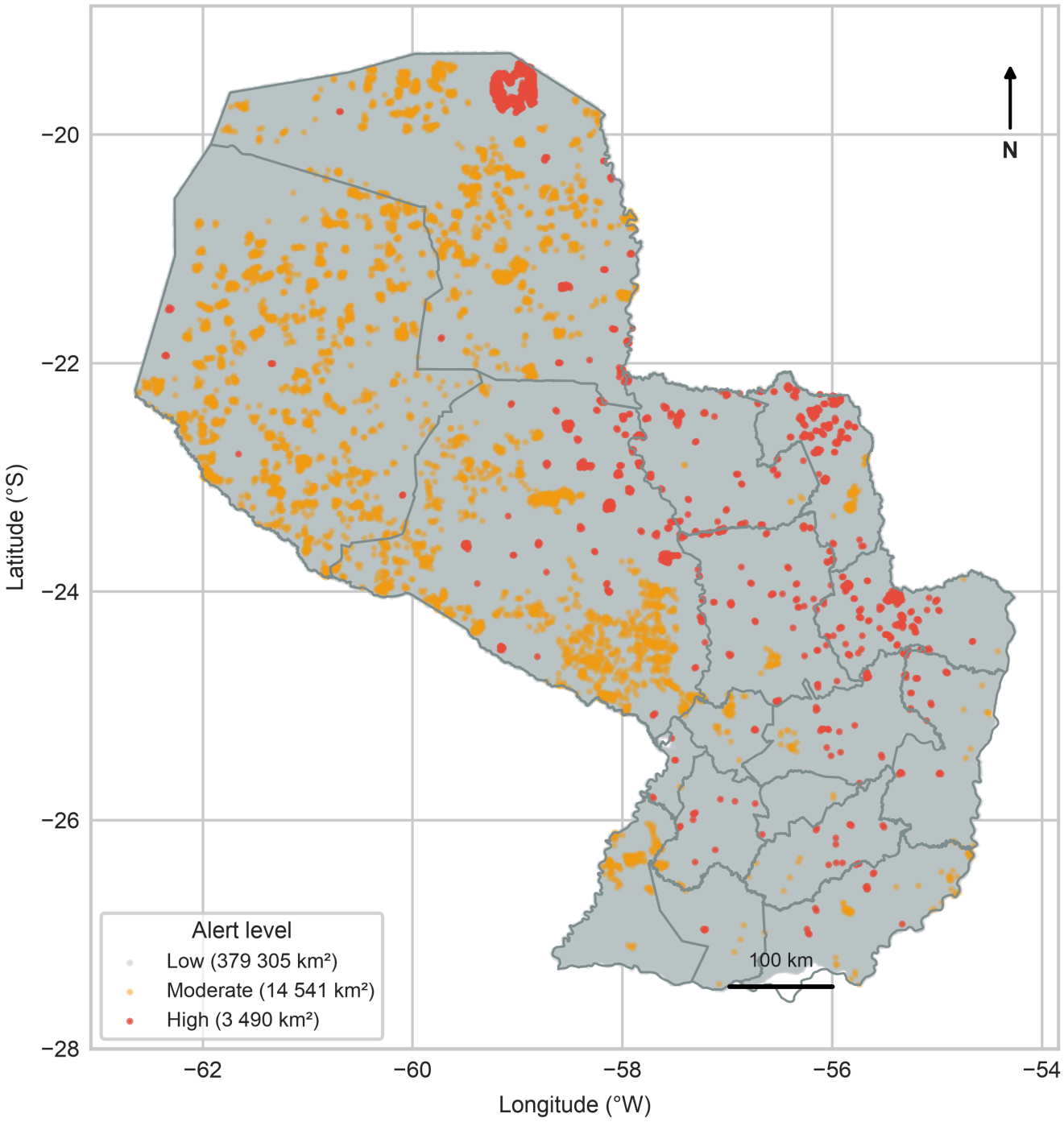


Figure 5. Illustrative national 1-km² alert map for 2024-09-09, showing the spatial distribution of low, moderate and high alert cells.

Figure S1.

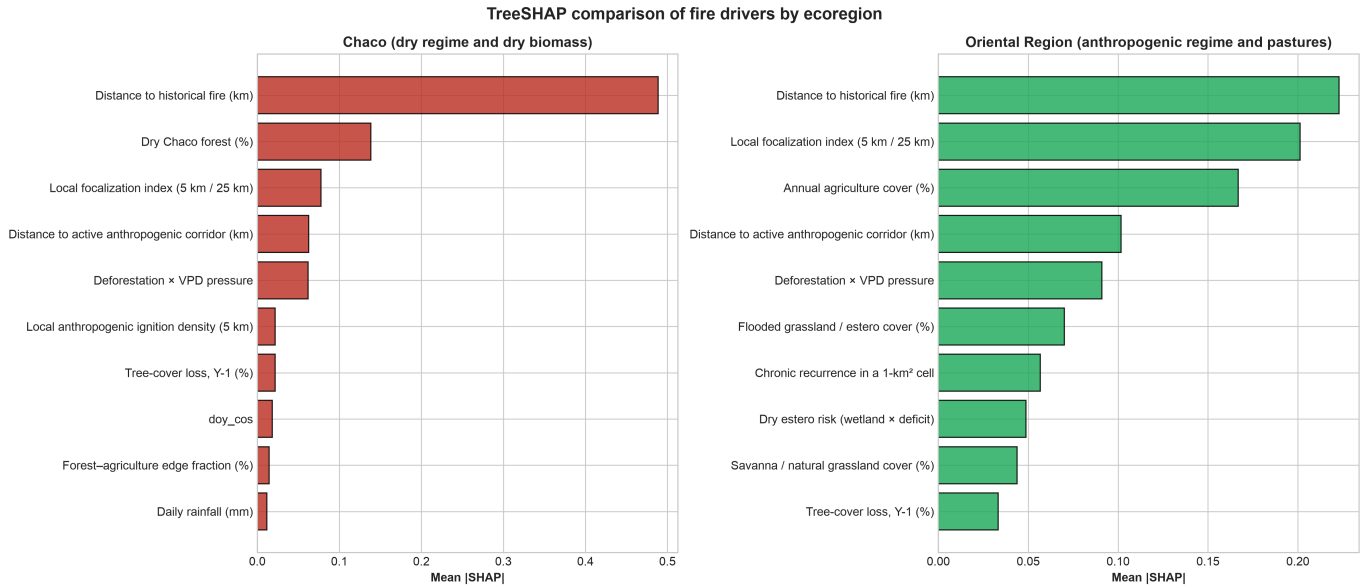


Figure S1. TreeSHAP comparison for the hybrid ignition model. The two regions use separate models and feature sets.

Figure S2.

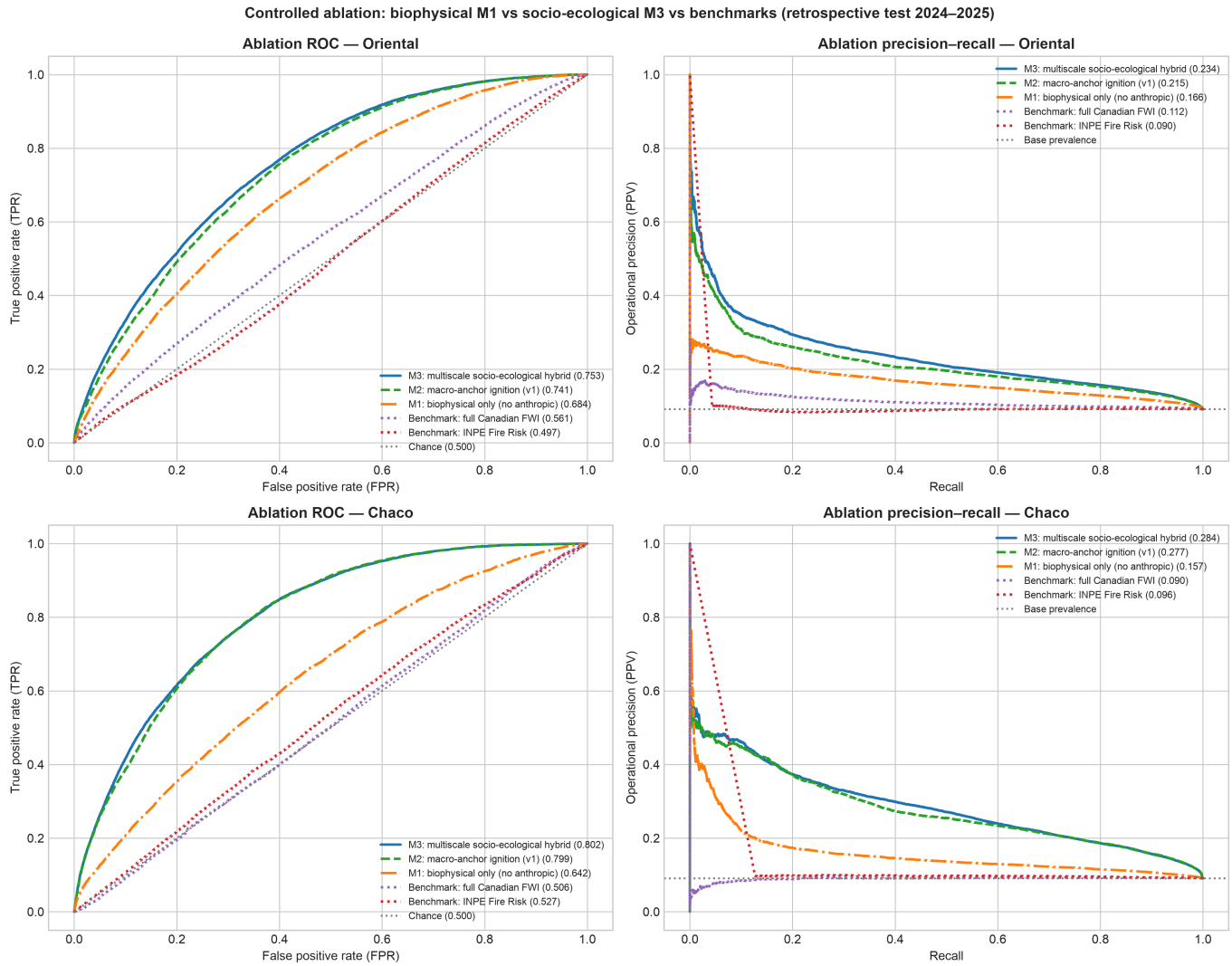


Figure S2. Controlled ablation under the D2 test. The figure complements Table S12.

Figure S3.

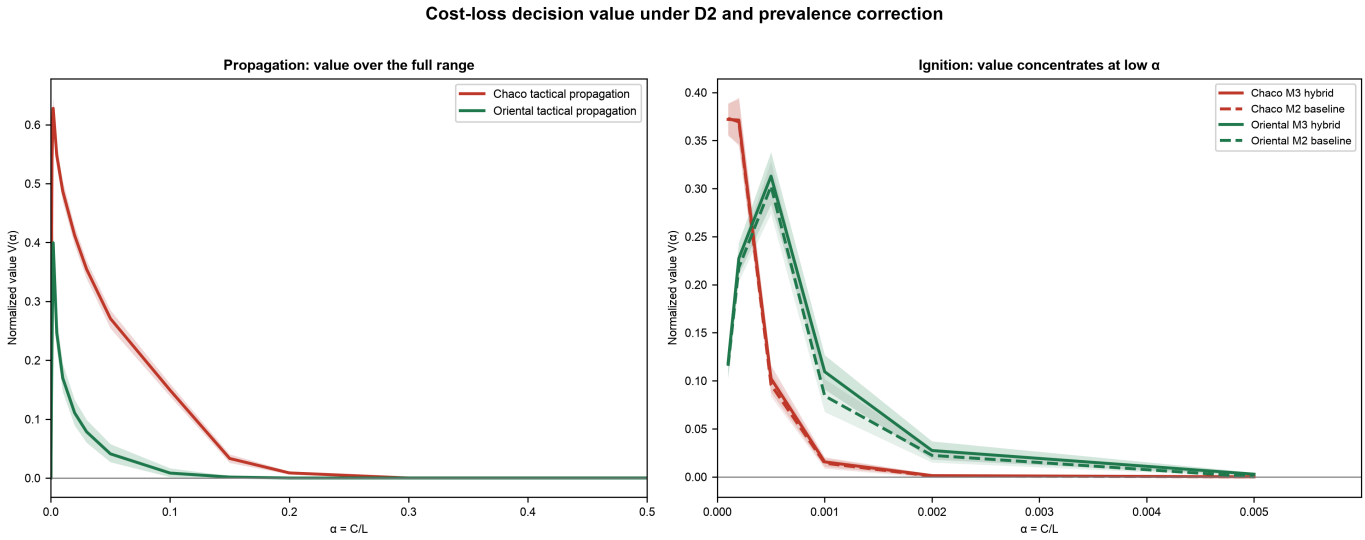


Figure S3. Cost-loss curves with 95 % date-grouped bootstrap bands (B = 1 000). The propagation panel uses the full α range; the ignition panel is truncated at α=0.006 because the decision value is zero or negligible at larger values. The figure complements Table S14.

Figure S4.

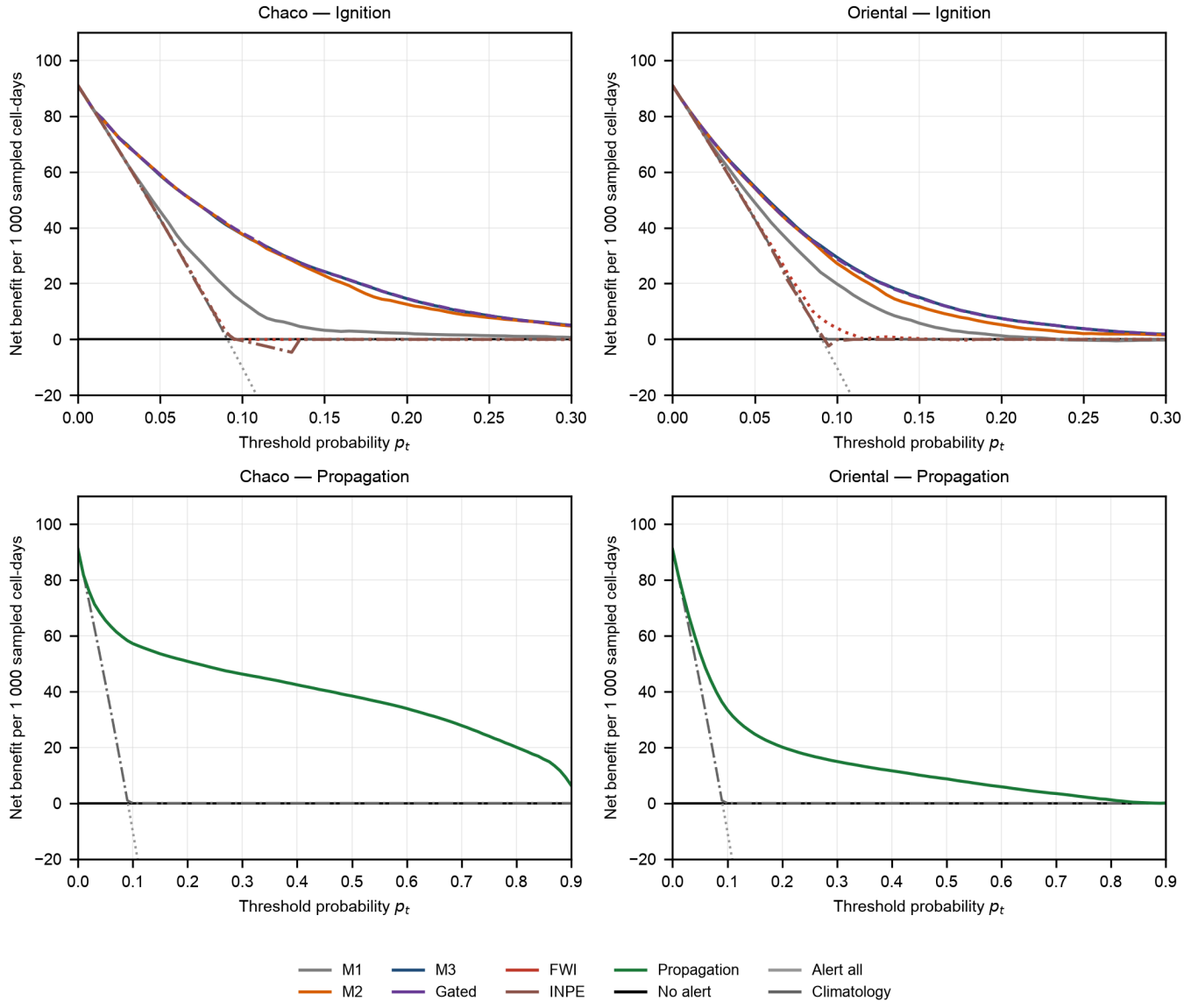


Figure S4. Decision curve analysis: net benefit per 1 000 sampled cell-days as a function of the threshold probability p_t, computed on the case-control test sample. Ignition (0–0.30) and propagation (0–0.90) are shown separately by region. References are no alert (NB = 0), alert all and climatology. FWI and INPE are uncalibrated indices; for each p_t their operating threshold was selected on 2023 validation to maximize net benefit and then frozen for the test. The y-axis is clipped at −20; the alert-all reference falls below it at high p_t.

Figure S5.

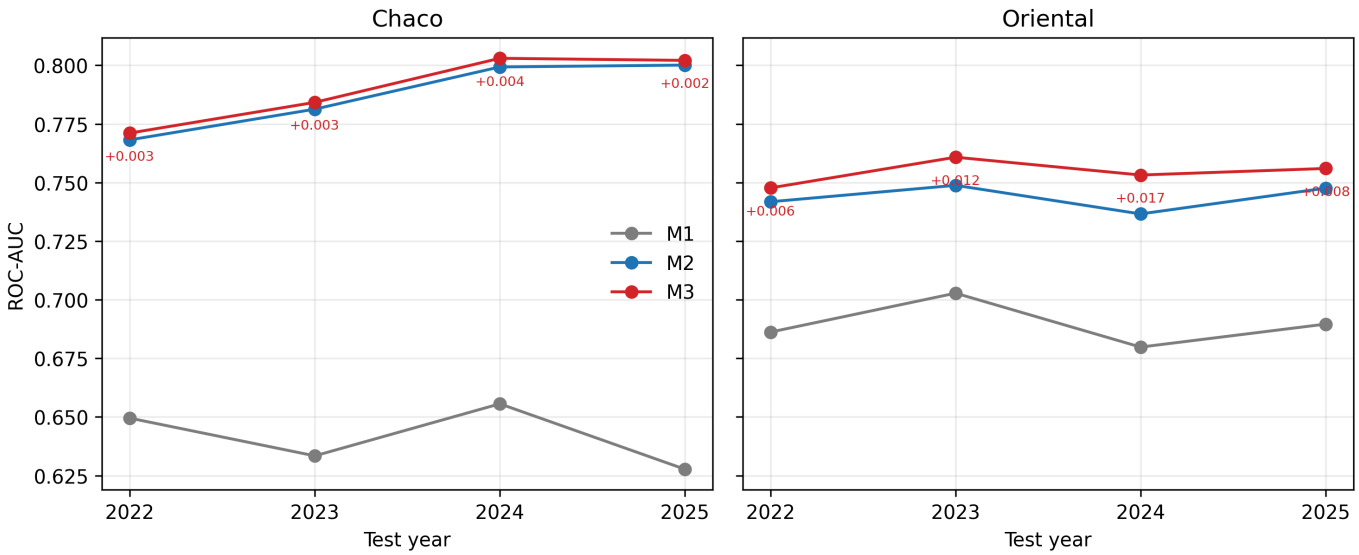


Figure S5. Rolling-origin ROC-AUC by test year for M1, M2 and M3. Numbers next to M3 are ΔROC(M3−M2).

Figure S6.

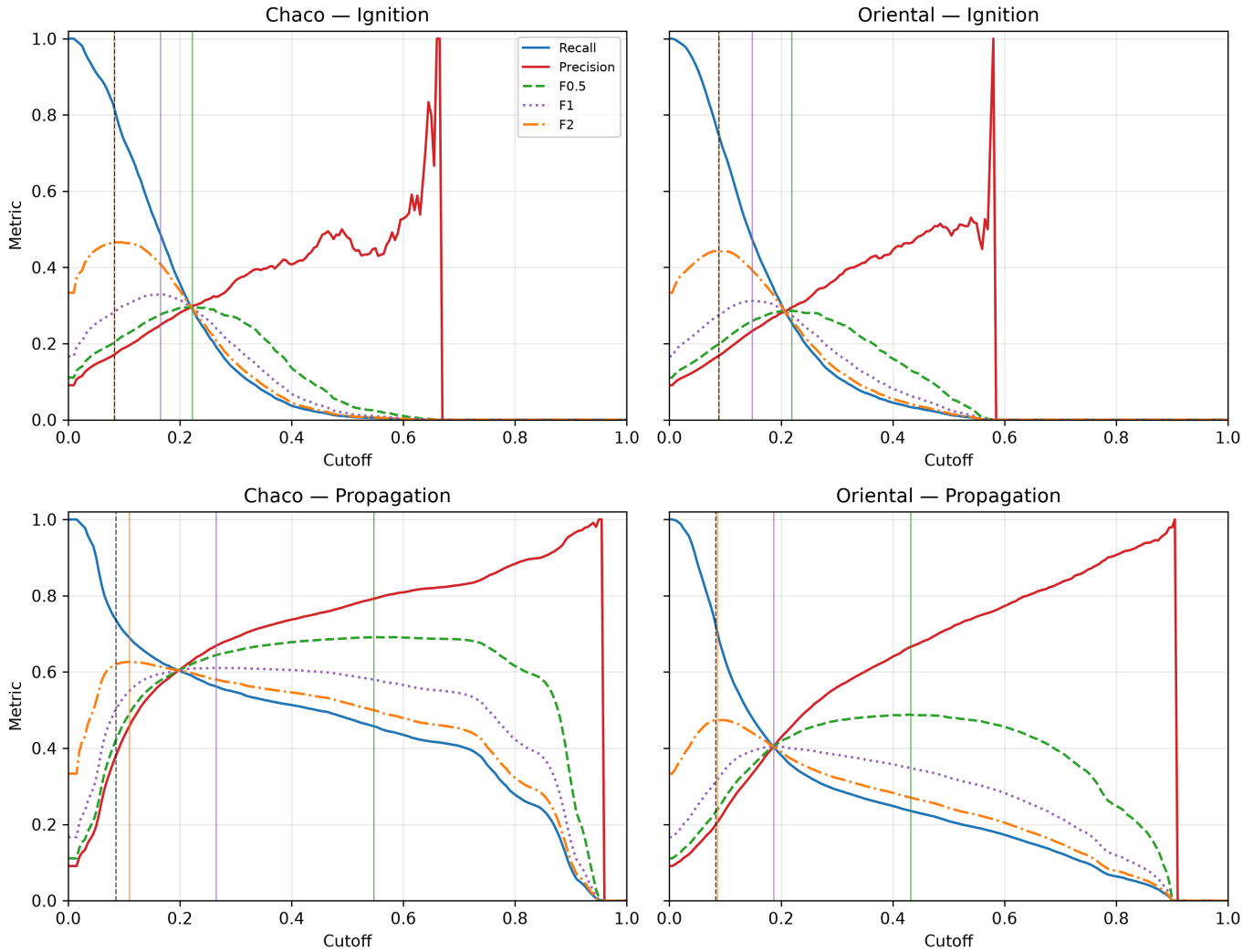


Figure S6. Validation precision, recall, F0.5, F1 and F2 versus cutoff for M3 ignition and the propagation model, by region. Vertical lines mark the F0.5, F1, F2 and Youden cutoffs; test metrics at those cutoffs are in Table S14 (continued).

Figure S7.

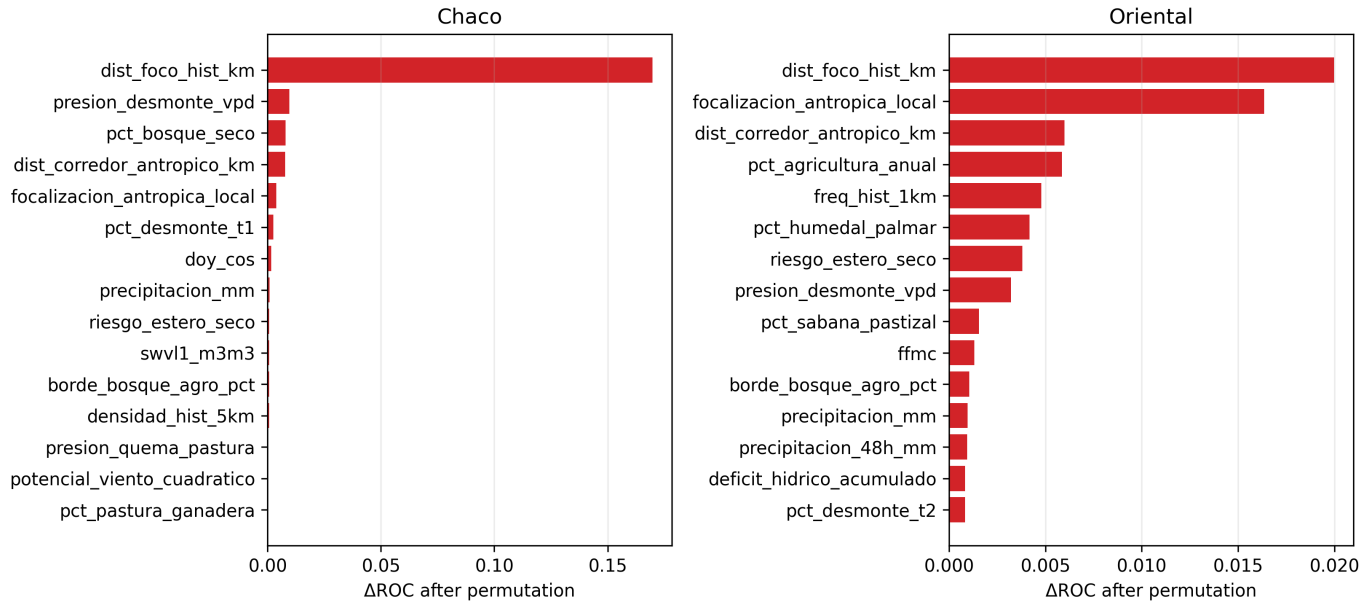


Figure S7. Permutation importance of the M3 features on the retrospective test (mean ΔROC after shuffling, three repeats), by region.

Figure S8.

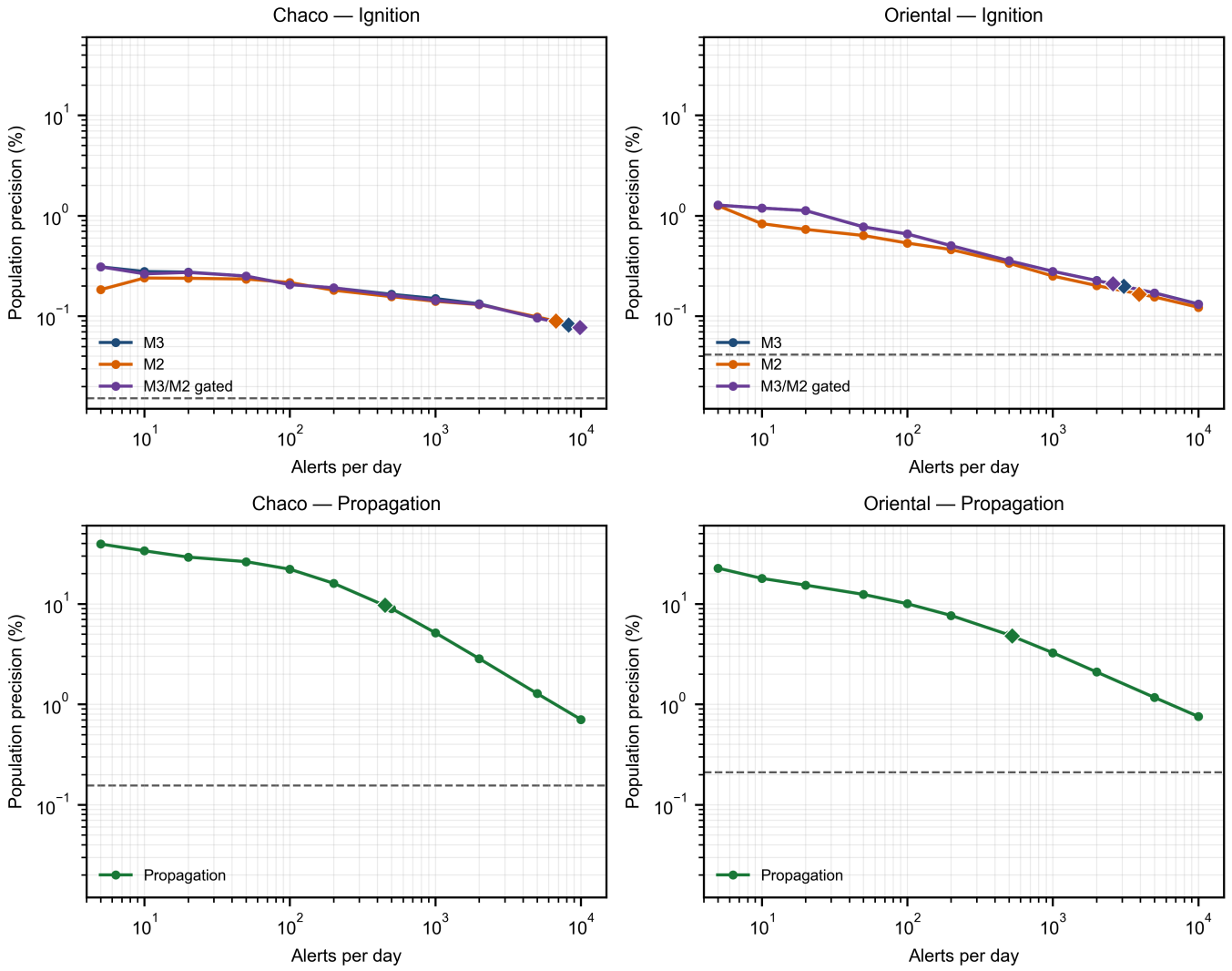


Figure S8. Population precision as a function of the daily alert budget. Curves sweep the population-corrected score threshold and report weighted precision at the resulting number of alerts per day; diamonds mark the frozen operating point. Ignition panels include M3, M2 and the gated policy; propagation panels show the tactical model. The dashed line is the population prevalence.

Figure S9.

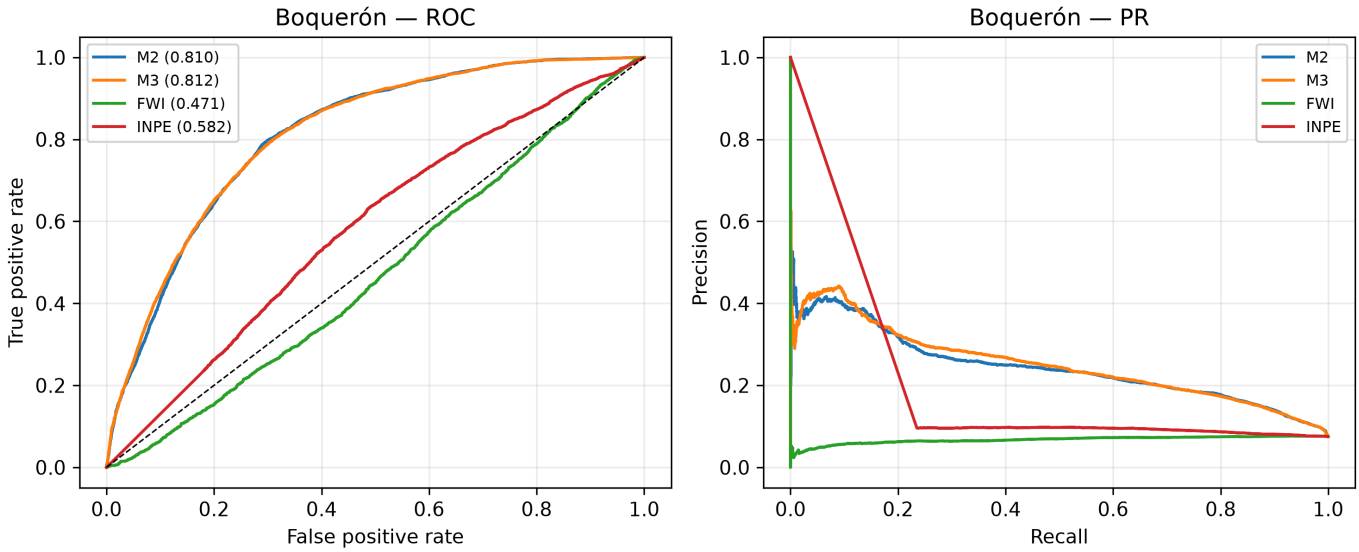


Figure S9. External validation in Boquerón: ROC and precision–recall curves for M2, M3, FWI and INPE on 2024–2025.

Figure S10.

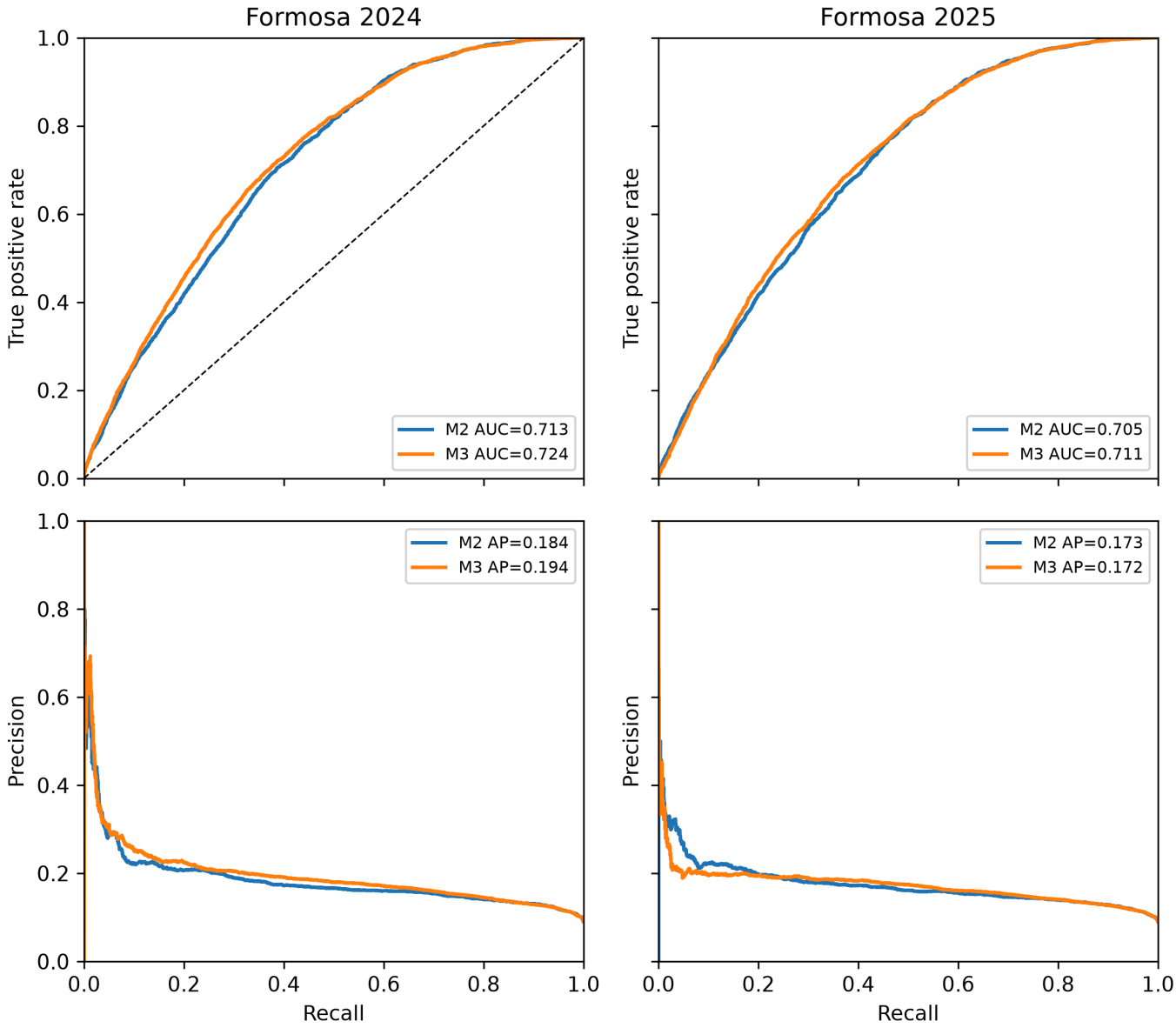


Figure S10. External cross-country validation in Formosa. ROC and precision–recall curves for G0-C in 2024 (left) and 2025 (right). M2 (blue) and M3 (orange) use frozen Chaco boosters with local Platt recalibration. The test is case-control (1:10), not population-weighted.